

Version #	Date	Details of change	Slide no.
2	June 2024	Diagram updated for clarity	15

Schedule management

Learning objective:

20. Understand schedule management as the ability to undertake time-based planning with an emphasis on activities and resource.

What is schedule management?

Time scheduling is used to develop schedules for when work is planned to be completed in projects.

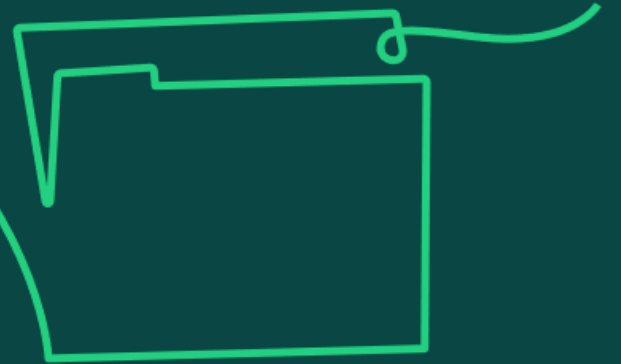
It's time-based planning with an emphasis on activities and resource.

It's the process for developing and maintaining schedules.

It shows when work for a particular project is planned to be done.

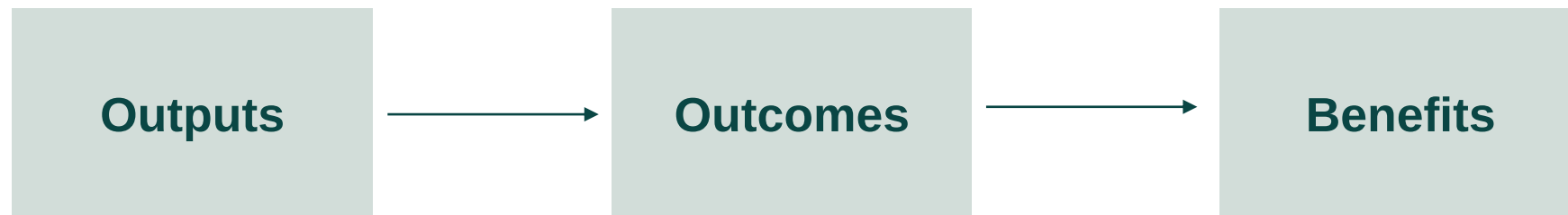
It considers dependencies and can be for internal and/or external resources and activities.

20a) Know how to define scope in terms of outputs, outcomes and benefits (including use of product, work and cost breakdown structures).



Defining scope

- Scope is defined by the outputs (and the work needed to create them), outcomes and benefits being delivered. Typically, a high-level scope will be recorded in the business case.
- A project's outputs are the tangible or intangible deliverables or products being created which produce the outcomes of changed circumstances, behaviours or practices, to create the benefits in line with the defined acceptance criteria.

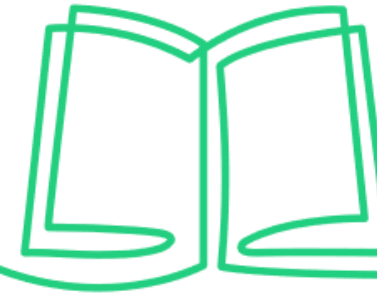


Defining scope in detail

- It's important to be clear about the boundaries and interfaces with adjacent projects to avoid duplication, conflicts or omission of work within projects or business-as-usual work.
- The detailed scope of work emerges from the breakdown of the chosen option to meet the sponsor's requirements.
- Once a solution has been identified to meet these requirements, breakdown structures can be used to illustrate what products and activities are needed to deliver the project.
- For projects using a linear life cycle, breakdown structures provide a hierarchical view of what is needed to deliver the project. Iterative lifecycle projects will often use a Product Backlog (essentially a prioritized list of what is needed).
- Content created at each level supports the creation of the next level of the hierarchy until the project deliverables are identified. The same can then be done for the detailed 'workpackages needed to create the products.
- This enables decisions to be made about who in the project will take responsibility for carrying out aspects of work, supervising activity and reporting on progress.

Breakdown structures

Illustrating what needs to be done



Product breakdown structure (PBS)

A PBS identifies the scope of a project to be delivered.

The end output of the project is placed at the top level, defining all components moving down levels until reaching the individual products to be delivered.

Each product has defined acceptance criteria and quality control methods.

This can be used to secure stakeholder agreement of the deliverables and agree what is in and out of scope.

Work breakdown structure (WBS)

A WBS defines a baseline scope of work capturing the activities or work to be completed to deliver the project requirements.

The lowest level of detail identified is a work package.

The work required and skill set needed to complete can be reviewed, and work requiring a similar skill set can then be grouped. This supports allocation of resources, estimating and scheduling of the work packages.

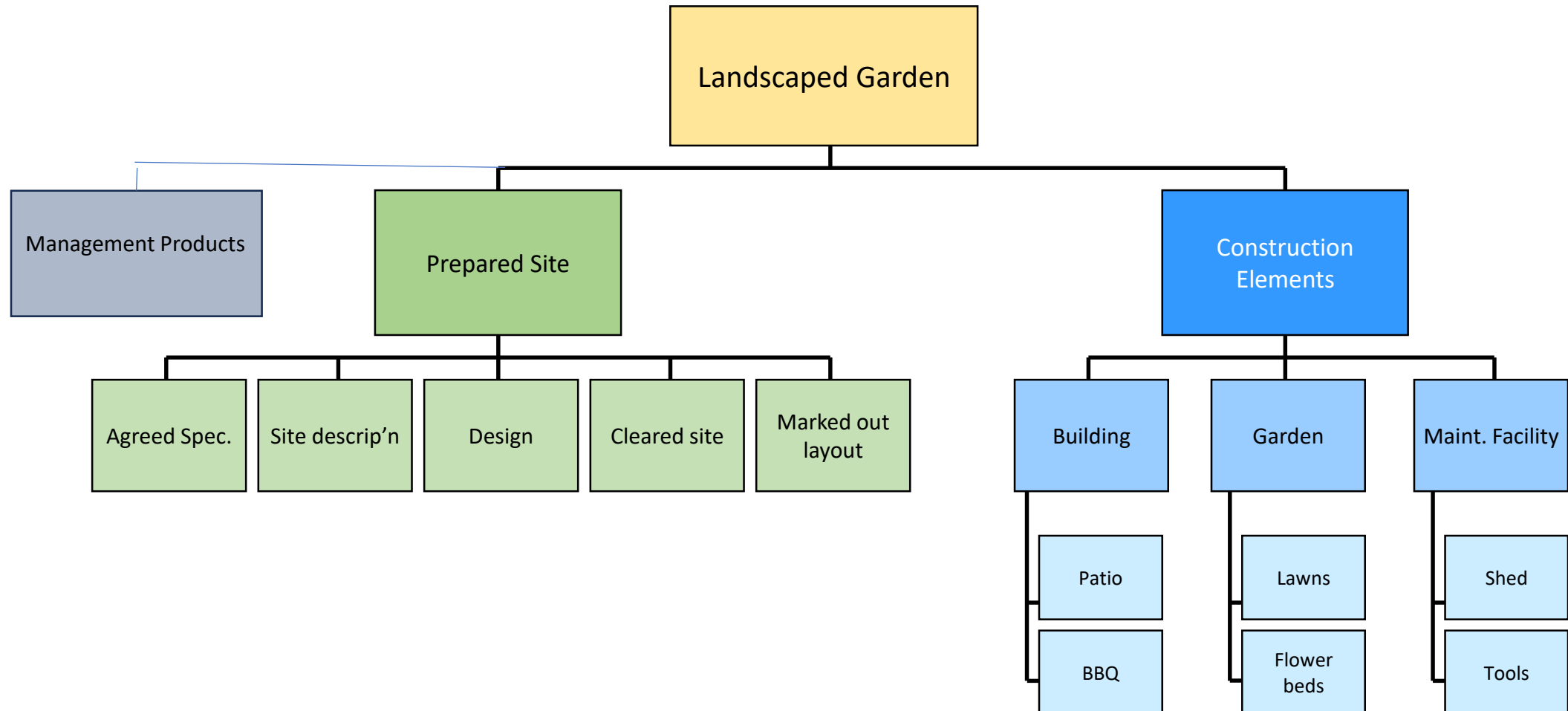
Cost breakdown structure (CBS)/ Organisational breakdown structure (OBS)

A CBS provides a financial view by breaking down the project into cost elements.

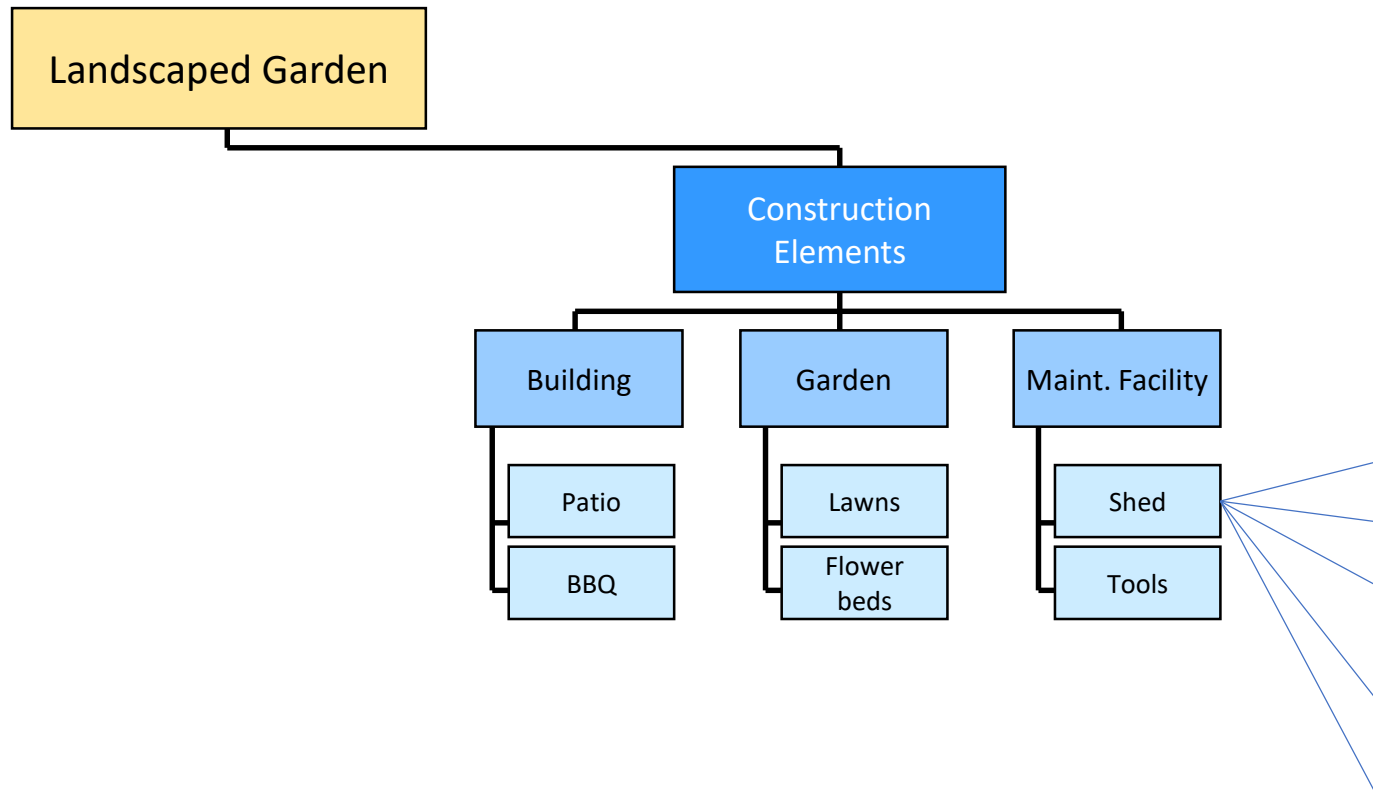
It can be used to show either the costs assigned to work packages, using the WBS, such as people, equipment, materials or other required resource, or the costs assigned to functional areas and third parties using the organisational breakdown structure (OBS).

This can be used to track project spend to compare against the estimates produced.

Product Breakdown Structure



Work Breakdown Structure



Workpackages for “Shed” Output

Digging out for pad

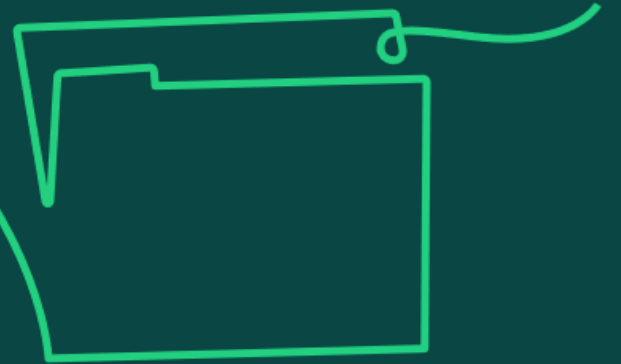
Pour concrete for pad

Build shed shell

Fit out shelves

Fit windows

20b) Understand there are links and dependencies between activities within a project and business-as-usual activities, for example business-as-usual activities, costs, quality, risks and scope can all impact the schedule for the project.



Links and dependencies

All the following can have an impact on a project and should be considered as part of schedule management.

Relationship between project tasks

The links between activities and their implications.

Scope definition

Prevent risk of delays through misunderstanding of what is in and out of scope for the project.

Dependencies

Between different project activities and business as usual (BAU).

Stakeholders

From a technical and BAU perspective to inform on potential impacts to time / resource.

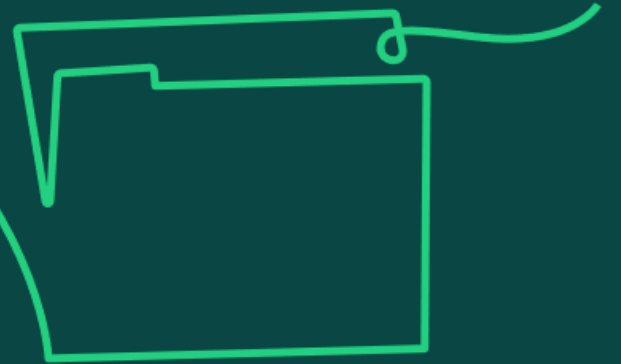
Business change/ Adoption

Activities that are resourced by BAU not the project.

Schedules Clashes

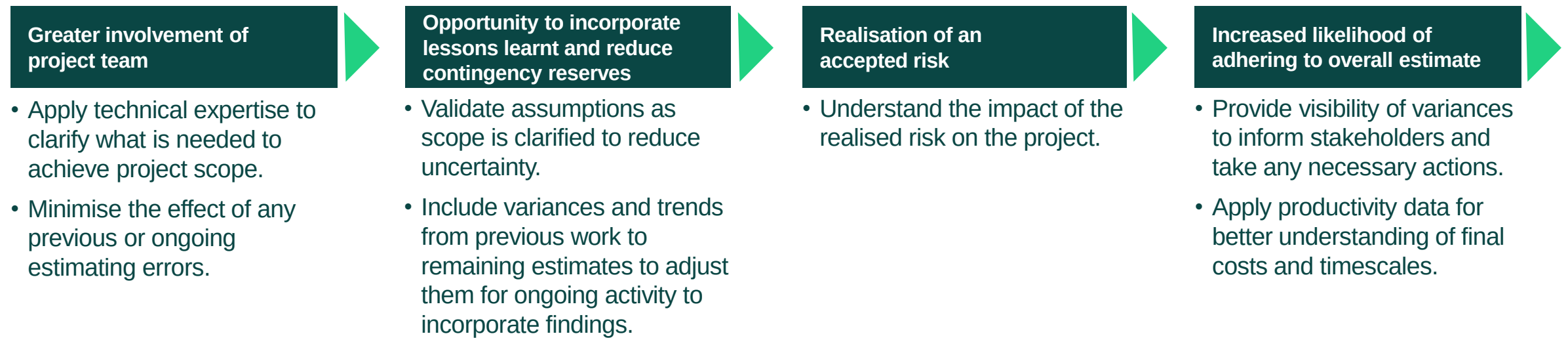
Advise of activities and any potential conflicts requiring resolution.

20c) Understand the reasons for and benefits of re-estimating, and schedule optimisation, throughout the project life cycle.



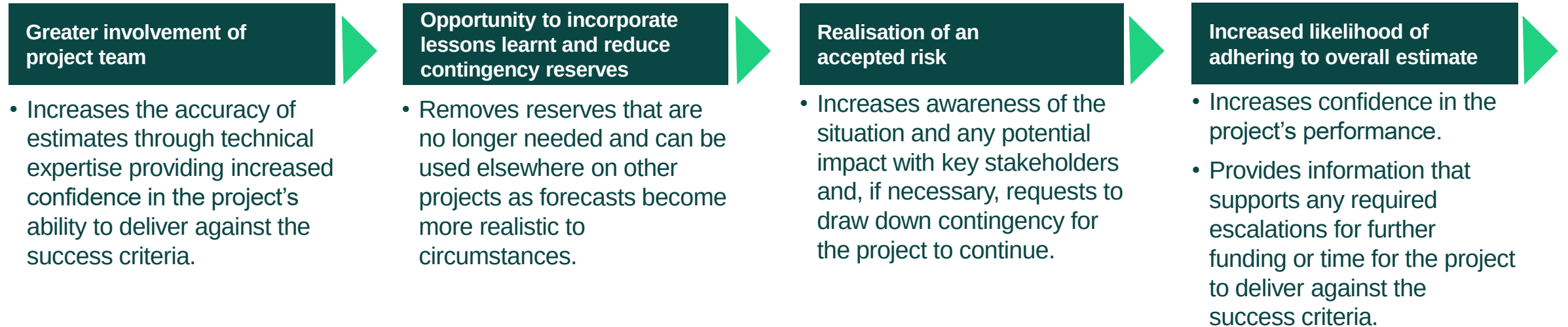
Why re-estimate and optimise?

An estimate is just that – an approximation of time to complete tasks. The activity of estimating tasks should not be a one-off but should be ongoing. As more information is known, re-estimating should happen. This will increase confidence in the estimates being used to create the schedule.



To fully optimise scheduling techniques, the concepts that feed into the activity must be understood. The WBS, estimating, dependencies, resource allocation, alignment of success criteria and baselining are the foundations for scheduling and for the optimisation of the schedule for a successful project.

The benefits of re-estimating

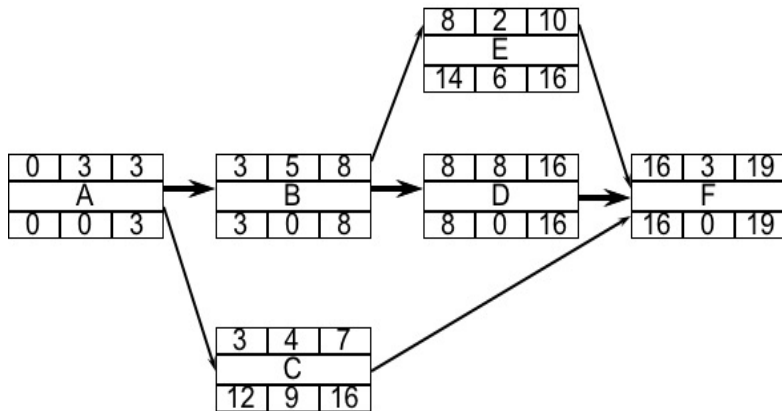


Critical path and Critical Chain

- Two of the main methods used to support scheduling optimisation are critical path and critical chain:
 - The **critical path** method places emphasis on the time required to complete all the individual activities in a logical order. Identifying the relationship between tasks, and the time estimate for each is essential in identifying the critical path. Resources can be used flexibly.
 - The **critical chain** method places emphasis on **resources**, attempting to keep them at a constant utilisation. It uses an overall Project Buffer, rather than flexibility in individual activities.
- The critical chain method, alternatively called the 'resource critical path', places the emphasis on the resources (labour and non-labour) in a project, while the **critical path** emphasises the activities & time taken.
- If resources were unlimited and always available, the critical path and critical chain method for scheduling would provide the same result.
- The results of these methods are usually presented as a visualisation, such as a Gantt chart that shows either activities or resources as bars on a timeline.

Critical Path

- Identifies the activities or tasks in the plan that if delayed will affect the end date.
- These critical activities, have no 'float' and create the longest pathway of activity on the project and therefore form the Critical Path.
- We can manage a critical path schedule by moving resource from non-critical to critical activities, utilising the available float.
- The start and finish of activities are the focus for critical path reporting.



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
A	Red	Red	Red																
B				Red	Red	Red	Red	Red											
C				Blue	Blue	Blue	Blue	Hashed	Hashed	Hashed	Hashed	Hashed	Hashed	Hashed	Hashed	Hashed			
D									Red	Red	Red	Red	Red	Red	Red	Red			
E									Blue	Blue	Hashed	Hashed	Hashed	Hashed	Hashed	Hashed			
F																	Red	Red	Red

Red = Critical Path; Blue = Non-critical activities; Hashed = Float in non-critical activities



Critical chain

- The critical chain method takes the most likely duration estimates for each activity using a set resource, this resource is for THAT activity and should not be moved to other activities.
- A common approach is to add up these estimates and an amount equal to half of this overall time is held as a project buffer at the end for the chain of activities rather than having a float for individual activities, as in Critical Path.
- The buffer is there to protect the whole of the project and extra time will be taken from it if the 'most likely' times are forecast/are exceeded.

